

Kevin Li

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Profile: Biostatistics M.S. graduate with hands-on experience in bioinformatics, clinical statistical modeling, and agentic coding-based data pipelines. Proficient in R, Python, SAS, and Linux, with a track record of conference presentations and award recognition across National Institutes of Health (NIH) and Stanford University settings. Proven ability to contribute effectively through independence and cross-functional collaboration, and eager to bring computational expertise to a team advancing data-driven implementations.

Skills

- **Statistical modeling:** survival analysis, regression, mixed-effects models, machine learning, and neural network methods
 - Received Course Certificate in **Unsupervised Learning, Recommenders, and Reinforcement Learning** from DeepLearning.AI at Stanford University via Coursera (Mar. 2026)
- **Large language model (LLM) application** for code optimization, data analysis interpretation, and automated information extraction pipelines
 - Received a Specialization Certificate in **Prompt Engineering** from Vanderbilt University via Coursera (Mar. 2026)
- **Proteomic and genomic data:** processing, curation, and analysis
- **Application programming interface (API) design** for data integration across research platforms
- **Programming:** R, SAS, Python, MATLAB, Linux, GitHub version control
 - SAS Base Programmer Certification (May 2025)
- **Regulatory compliance:** Good Clinical Practice (GCP) and Clinical Data Interchange Standards Consortium (CDISC)
 - NIH Clinical Trials Network - Good Clinical Practice Certificate (July 2025)

Education

University of California, San Diego

M.S., Biostatistics (Sept. 2023 - June 2025), GPA: 3.85

- Coursework in clinical trials, survival analysis, longitudinal data, statistical learning

University of California, Santa Barbara

B.S., Biochemistry and Molecular Biology (Sept. 2019 - June 2023), GPA: 3.5

- Relevant coursework in linear algebra, differential equations, and vector calculus

Work/Internship Experience

National Center for Advancing Translational Sciences (NIH), Rockville, MD

Postbaccalaureate Fellow, Sept. 2025 – Present

Independently developed an LLM-powered API in collaboration with Dr. Qian Zhu and the NIH Informatics Core, transforming unstructured natural history study data into machine-readable clinical datasets to advance rare disease therapy development (preprint submitted). Led cutting-edge application of graph neural networks to large-scale biomedical knowledge graphs, with findings positioned to inform drug repurposing strategies for underserved rare disease communities. Developed a pipeline that scrapes rare disease communities to identify patient-reported off-label drug use, supporting a data-driven drug repurposing framework for underserved rare disease communities.

Stanford University, Department of Medicine, Stanford, CA

Research Assistant (10-20 hours/week), Sept. 2023 - Sept. 2025

Full-time Intern, Summers 2023 & 2024

Architected automated Linux and R pipelines to process and analyze over 2,000 murine single-cell datasets sourced from public genomic databases with Dr. Paul Cheng, directly enabling investigations into nonsense-mediated decay dysregulation during transgenic mouse development. Drove single-cell RNA sequencing analysis of Lamin A/C expression across smooth muscle cell populations, generating key insights into the molecular mechanisms underlying progeria disease progression across age groups.

Stanford University, Department of Genetics, Stanford, CA**Summer Intern**, June - Sept. 2022

Developed R-based computational workflows for single-cell metabolomic analysis within the Dr. Michael Snyder lab, integrating tools for biological pathway visualization, metabolomic profiling, and enrichment analysis across the KEGGgraph, MetaboAnalystR, and FELLA frameworks to support multi-omics research.

Carnegie Institution for Science, Stanford, CA**Research Intern & Data Analysis Assistant**, June 2018 - Sept. 2021

Executed multi-faceted proteomics research across the Dr. Zhiyong Wang lab, leveraging mass spectrometry software to analyze complex proteome datasets and deliver critical findings to senior researchers. Spearheaded the deployment of upgraded Thermo Proteome Discoverer software to advance quantification of protein expression profiles in plant disease models, while independently conducting mutant phenotype characterization and genotyping to drive investigations into BAM8 protein function in *Arabidopsis*.

Peer-Reviewed Publications

1. Li K., Sid E., Zhu Q. (2026). LLM-enabled Natural History Study Analysis to Support Rare Disease Research. medRxiv. <https://doi.org/10.64898/2026.09.17.26363303>.
2. Kim T., Park C. H., Hsu C., Kim Y., Ko Y., Zhang Z., Zhu J., Hsiao Y., Branon T., Kaasik K., Saldivar E., Li K., Pasha A., Provart N. J., Burlingame A. L., Xu S., Ting A. Y., Wang Z. (2023). Mapping the signaling network of BIN2 kinase using TurboID-mediated biotin labeling and phosphoproteomics. *The Plant Cell*, 35(3), 975–993.
3. Li D. Y., Kundu S., Gu W., Cheng P., Jackson W. R., Zhao Q., Nguyen T., Ramste M. O., Monteiro J. P., Li K., Kundaje A., Wirka R., Quertermous T. (2024). A Multiomic Timecourse Atlas of Murine Vascular Disease Smooth Muscle Cell Transition Phenotypes and Their Perturbation with Tcf21 Knockout. *Arteriosclerosis, Thrombosis, and Vascular Biology* (Vol. 44, Issue Suppl. 1)

Presentations & Posters

1. Li K., Sid E., Zhu Q. "LLM-enabled Natural History Study Analysis to Support Rare Disease Research". AMIA Amplify Informatics Conference. May 18-21, 2026. Denver, CO. (Accepted for poster presentation)
2. Li K., Sid E., Zhu Q. "LLM-enabled natural history study analysis to support rare disease research." NIH Rare Disease Day, Feb. 27, 2026, Bethesda, MD.
3. Li K., Li D. Y., Cheng P. "Inefficient Nonsense Mediated Decay in Murine Cre/Lox-Mediated Conditional Knockouts." Biostatistics Symposium of Southern California. Feb 21-22, 2025. Irvine, CA.
4. Li K., Li D. Y., Cheng P. "Inefficient Nonsense Mediated Decay in Murine Cre/Lox-Mediated Conditional Knockouts." Cardiovascular Research Symposium. Aug 15, 2024. Stanford, CA.
5. Li K., Jason E., Qian H., Huang Y. "Solitude's Shadow: The Mental Health Maze in COVID Effects of Social Isolation During the COVID-19 Pandemic on Mental Health." UCSD Public Health Research Day. May 8, 2024. La Jolla, CA.
6. Li K., Webb M., Wang Z., Xu S. "A tale of two programs: a software comparison for label-free quantification." Intern Presentation Day. Aug 10, 2018. Stanford, CA.

Awards

- **Honorable Mention Presentation Award**, Biostatistics Symposium of Southern California, Feb 21-22, 2025, Irvine, CA for "Inefficient Nonsense Mediated Decay in Murine Cre/Lox-Mediated Conditional Knockouts."

Professional Memberships

- American Statistical Association (ASA)
- American Medical Informatics Association (AMIA)

I am a U.S. citizen born in Indianapolis, IN.